Table 2.5 The compression table created from the Huffman tree in figure 2.15

Character	Bit Sequence	Frequency
A	0	0.6
B	10	0.2
C	1100	0.07
D	1101	0.06
E	1110	0.05
F	1111	0.02

Looking at the sequences, the most important property is that they form a prefix code: no sequence is the prefix of another sequence in the code.

This property is the key point for the decoding: iterating on the compressed text, we immediately know how to break it into characters.

For instance, in the compressed text `1001101`, if we start from the first character, we can immediately see the sequence `10` that matches `B`, then the next `0` matches `A`, and finally `1101` matches `D`, so the compressed bit sequence is translated into “`BAD`”.

2.9 Analysis of branching factor[20]